BRAIN-AGE FAIRNESS & XAI PROJECT · TEAM HANDOVER

Right for the Wrong Reasons?
The whole project, explained from zero.

What we built, why, every result in plain words, where everything lives, and how to run it. No prior knowledge needed. Read it top to bottom once and you will know the project end to end.

3.71 yrs	None	−5.1 yrs	91%
average error on unseen brains	difference between men and women	IOP hospital brains look younger	hospital read from the model's insides

THE ANSWER IN ONE LINE

The model is accurate and fair across sex, but it is **partly right for the wrong reasons**: it secretly recognises the scanner, and its answer moves when only the scan's look changes.

Contents

1. The big idea
2. Words you need
3. The data
4. The model
5. The journey, step by step
6. Results 1–3: accuracy, fairness, unseen sites
7. Results 4: the heatmaps
8. Results 5–7: right for the wrong reasons?
9. The deliverables
10. The rules we followed
11. Where everything is & how to run it
12. Questions people will ask

1 The big idea

A **brain-age model** is an AI that looks at a brain MRI scan and guesses the person's age. Doctors and researchers care about the difference between the guess and the real age: if your brain "looks" 8 years older than you are, that might say something about brain health.

Most projects stop at "how accurate is it?". We asked two harder questions:

- **Is it fair?** Is it equally right for men and women, and for people scanned at different hospitals?
- **Is it right for the right reasons?** Is it actually reading the brain, or is it partly noticing *which machine* took the picture?

> **ANALOGY**
> A dog trained to find drugs in suitcases might really be smelling the plastic wrap one airport uses. It would be "right", but for the wrong reason. We checked whether our model is doing something similar with hospital scanners.

2 Words you need (glossary)

TERM	MEANING IN PLAIN WORDS
MRI / T1 scan	A 3D picture of the brain. T1 is the standard type showing tissue structure.
Brain-age gap (BAG)	Predicted age minus real age. Positive = brain looks older; negative = looks younger.
MAE	Mean absolute error: the average number of years the guess is off.
r (correlation)	How closely predictions follow real ages. 1.0 is perfect.
Train / validation / test	Train = the model learns from it. Validation = we tune and choose settings. Test = the final exam, opened **once**.
Fine-tuning	Taking an already-trained model and training it a bit more on our data.
Zero-shot	Using a model as-is, without training it on our data.
Site	The hospital (and its scanner) where a scan was taken. Ours: Guys, HH, IOP.
95% CI	Confidence interval: the range the true value very likely lies in. If it crosses 0, there is no clear difference.
Heatmap / attribution	A colour map showing which brain areas the prediction is most sensitive to.
Bias correction	A standard fix: brain-age models over-guess young people and under-guess old people.
Harmonisation (ComBat)	A standard method that removes each hospital's average "shift" from data.

3 The data

- **IXI dataset**: public brain MRI scans of healthy adults aged 20–86 from three London hospitals: **Guys** (Philips 1.5T), **HH** (Philips 3T), **IOP** (GE 1.5T).
- **485 people** after cleaning, split **339 train / 73 validation / 73 test**, the same split for both of our pipelines.
- 14 scans were excluded: 10 had no record, 2 had no age, 2 had conflicting records.

> **IMPORTANT EARLY FIX**
> The first version of our demographics table attached ages to the **wrong people** (a row number was used as the subject ID). Correct models looked broken. We rebuilt every label from the official `IXI.xls`. Both of us found this independently, and the public iAudit project agrees with the corrected labels. Everything in this document uses the fixed labels.

> **NOTE**
> IOP recruited younger people (average 41 vs about 48 at the other two). That is why every comparison is **adjusted for age**.

4 The model

SFCN (Simple Fully Convolutional Network, Peng et al. 2021) is a 3D neural network, pretrained on about 14,000 UK Biobank brains. It won a brain-age competition.

1. The scan is cleaned: skull removed, brightness unevenness corrected, aligned to a standard brain shape, cut to 160×192×160 voxels.
2. Six layers of filters find patterns, from edges up to "how big are the fluid spaces".
3. Everything is squeezed into **64 numbers**, the model's summary ("notes") of the brain.
4. A final layer turns those into a **probability for each age** (18 to 90). The predicted age is the weighted average.

The original could only output ages 42–82, so we gave it a new 18–90 output layer and fine-tuned it in two stages (first the new layer, then the whole network gently). DeepBrainNet and a second SFCN, built in the `experiments/` pipeline, serve as independent comparisons.

5 The journey, step by step

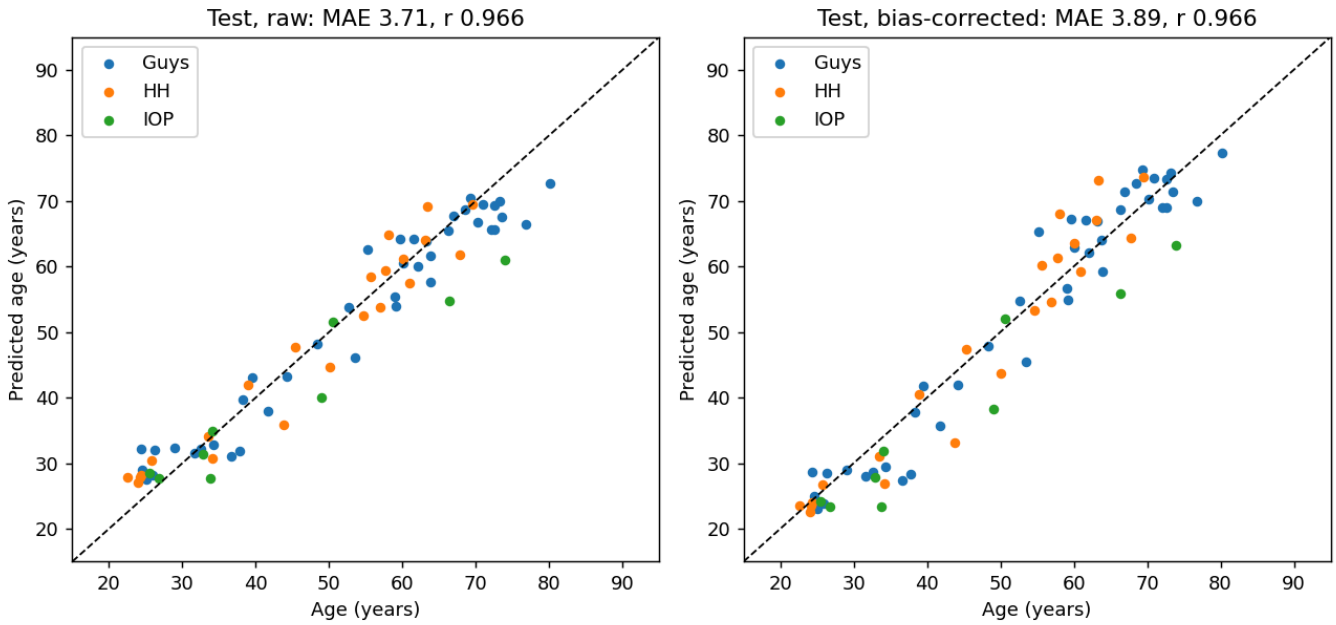
Labels fixed → Preprocess scans → Zero-shot check → Fine-tune → Test once → Fairness audit → Unseen-site test → Heatmaps → Wrong-reasons investigation → Website · passport · report

PHASE	WHAT WE DID	SCRIPT
1–2	Rebuilt labels; preprocessed all scans	`01b`, `02b`
3	Tested the original model without training	`03b`
4	Fine-tuned SFCN (Colab GPU)	`04`
5	One-time test evaluation + bias correction	`05`
6	Fairness audit (sex, site) + leave-one-site-out	`06`, `06b`
7	Heatmaps + brain-region analysis	`07`
9a–g	Hospital probe, scanner-look test, sharpness, blur, harmonisation, uncertainty, heatmap deletion test	`08` – `14`
—	Website, README visuals	`15`, `16`

6 Results 1–3: accuracy, fairness, unseen sites

① It is accurate

Before fine-tuning the error was **10.2 years** (young people were guessed at about 55). After fine-tuning: **3.71 years** on the test set (95% CI 3.10–4.38), correlation **0.966**. Published models on small datasets like IXI usually get 3–5 years.



Predicted vs real age on the test set. Dots near the diagonal line = good predictions.

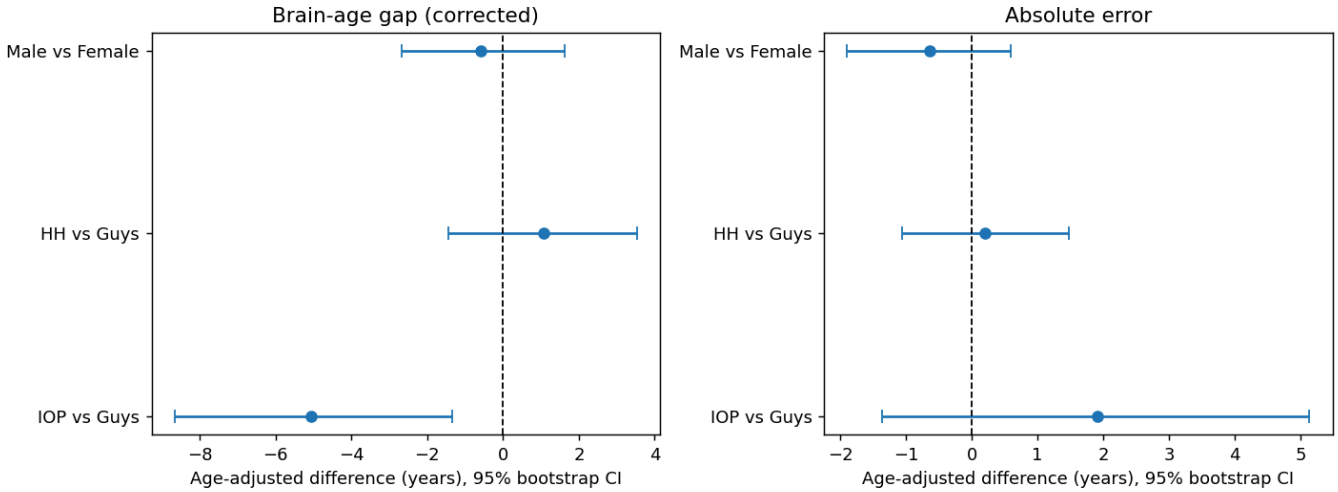
Comparison: on validation, our SFCN 3.90, the second SFCN 3.88, DeepBrainNet 4.12. The two SFCNs are equally good; test numbers must not be used to rank pipelines.

② Fair across sex, but one hospital stands out

COMPARISON (AGE-ADJUSTED)	BRAIN-AGE GAP	MEANING
Men vs women	−0.59 [−2.67, +1.64]	No difference ✅
HH vs Guys	+1.06 [−1.44, +3.53]	No difference ✅
IOP vs Guys	**−5.06 [−8.66, −1.34]**	IOP brains look ~5 yrs younger ⚠️

IS THE IOP FINDING REAL?

Yes. 8 of 9 IOP test people have a negative gap; removing any one person keeps it between −4.2 and −5.9; it appears separately in validation (−4.1) and test (−5.1). In training people it is absent (−1.0), so it is a problem with **new** IOP people.



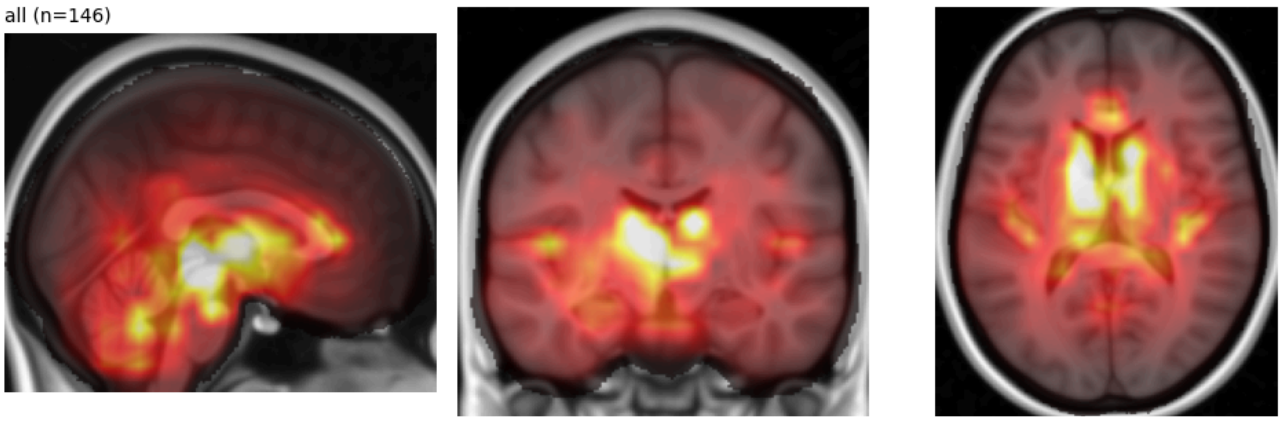
Group differences with confidence ranges. A range crossing the dashed 0 line = no clear difference.

③ A model that never saw IOP stumbles there

We trained three extra models, each with one hospital left out. Without IOP, error on IOP rose by **3.3 years** (6.8 vs 3.6). Without HH, nothing changed. IOP is the hard hospital to generalise to.

7 Results 4: the heatmaps

Heatmaps show where the prediction is most sensitive. Ours light up the deep brain areas around the fluid spaces (ventricles), which are known to change with age.



Average heatmap over 146 held-out brains.

Method note: standard Grad-CAM gave empty maps for 20–80% of people, so we used gradient × activation (HiResCAM-style). This choice was made before looking at any group results.

TWO TRUST TESTS, BOTH PASSED

Sanity check: scramble the model's layers and the heatmaps change completely (similarity only 0.13), so they come from what it learned. **Deletion test:** blank out the 5% of brain the heatmap ranks highest and the prediction moves **4.8 years**; a random 5% moves it only 1.0; the lowest-ranked 5% only 0.5.

Heatmap patterns differ between hospitals, but **not between men and women**.

8 Results 5–7: right for the wrong reasons?

⑤ The model learned the scanner

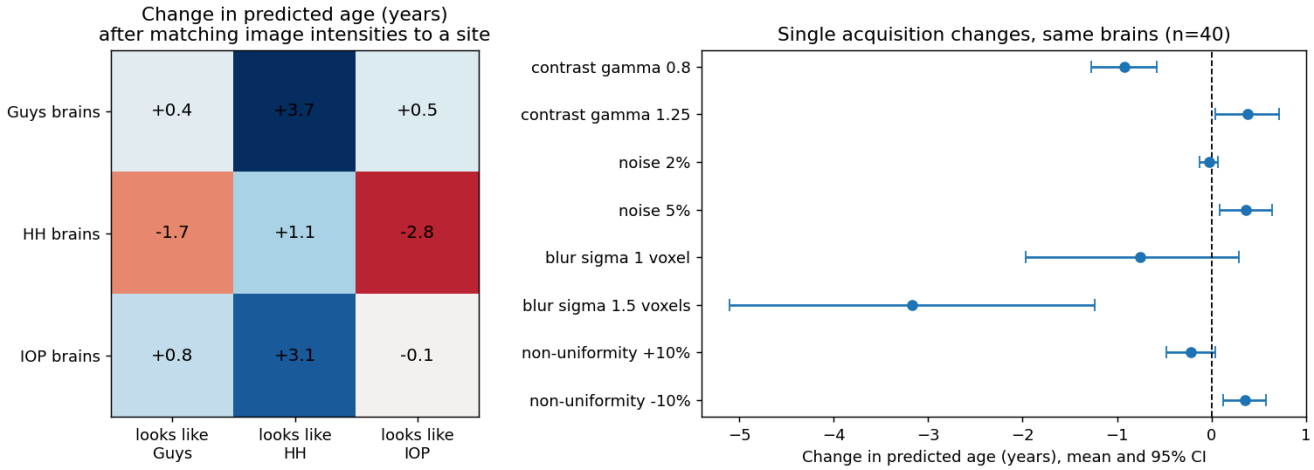
We took the model's 64 internal numbers and trained a simple program to guess the hospital from them alone.

WHAT THE GUESSER COULD SEE	HOSPITAL GUESSED CORRECTLY (CHANCE 33%)
Only age and sex	41%
The model's 64 numbers	94%
The 64 numbers, age and sex removed	**91%**

The model was **never told** about hospitals, yet its "notes" carry a scanner fingerprint. It was already there in the original UK Biobank model (93%).

The scanner-look test (the key experiment)

We took the **same brain** and changed only its brightness pattern to match another hospital's typical scan. The anatomy never changed.



Left: average change in predicted age when each hospital's brains get another hospital's look. Right: single changes (contrast, noise, blur).

- HH brains given IOP's look: **3.9 years younger**, all 45 of 45.
- IOP brains given HH's look: **3.1 years older**, all 18 of 18.

WHAT THIS PROVES

The same brain gets a different age depending on how it was scanned. The model is **partly right for the wrong reasons**.

⑥ But IOP's 5 years are not a simple scanner trick

SUSPECT	TEST	RESULT
Brightness	Swap Guys ↔ IOP brightness (age-adjusted)	≤ 1 year ❌
Blur	IOP scans are much blurrier at the same resolution; blur other brains to IOP's level	Not younger (+0.9 / 0.0) ❌
The fingerprint	Remove it with ComBat (hospital recognition 91% → 36%, accuracy unchanged)	Offset stays (−4.5 → −4.0) ❌

A CAREFUL CHECK THAT CHANGED A CONCLUSION

Because IOP's volunteers were younger, IOP's "typical look" was first built partly from younger brains. Once we adjusted each hospital's look for age and sex, the apparent IOP effect almost vanished. This is why we always adjust for age.

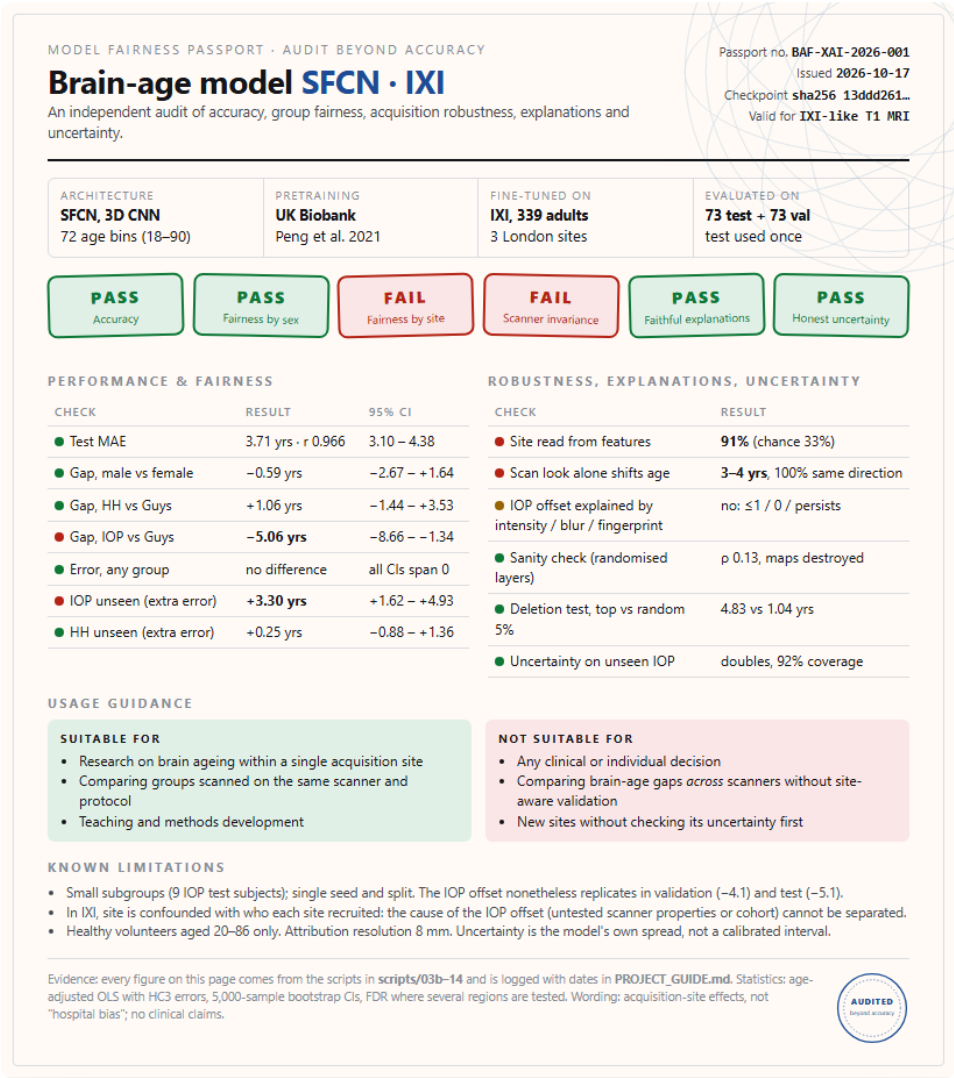
So what causes it? Either scanner properties we did not test, or IOP's volunteers genuinely differ. IXI cannot separate the two; that would need the same people scanned at several hospitals.

⑦ It knows when it is unsure

The model gives a probability for every age, so it also shows how sure it is. It is **more unsure about IOP** (+2.5 years), and **twice as unsure** when it never saw IOP (12.5 vs 6.1), while its "90% range" still contains the true age 92% of the time. It is less accurate there but admits it: **not confidently wrong**. For single scans the signal is weak; it is a group-level warning.

9 The deliverables

WHAT	WHERE	WHAT IT IS
Interactive website	`demo/index.html`	Story in chapters with interactive charts + "explore a brain" (pick a brain, toggle hospital looks). Works offline; just double-click.
Fairness passport	`demo/passport.pdf`	One-page certificate: pass/fail stamps, evidence, what the model is and is not suitable for.
Report draft	`report/report_draft.md`	Full academic write-up. **Team contributions section still to fill.**
README	GitHub front page	Animated banner, "same brain, three looks" animation, scorecard, seven findings.
Project log	`PROJECT_GUIDE.md`	Every result with its date, method and caveats.



MODEL FAIRNESS PASSPORT · AUDIT BEYOND ACCURACY

Brain-age model SFCN · IXI

An independent audit of accuracy, group fairness, acquisition robustness, explanations and uncertainty.

Passport no. BAF-XAI-2026-001
Issued 2026-10-17
Checkpoint sha256 13ddd261…
Valid for IXI-like T1 MRI

ARCHITECTURE	PRETRAINING	FINE-TUNED ON	EVALUATED ON
SFCN, 3D CNN 72 age bins (18–90)	**UK Biobank** Peng et al. 2021	**IXI, 339 adults** 3 London sites	**73 test + 73 val** test used once

PASS Accuracy · PASS Fairness by sex · FAIL Fairness by site · FAIL Scanner invariance · PASS Faithful explanations · PASS Honest uncertainty

PERFORMANCE & FAIRNESS

CHECK	RESULT	95% CI
Test MAE	3.71 yrs · r 0.966	3.10 – 4.38
Gap, male vs female	−0.59 yrs	−2.67 – +1.64
Gap, HH vs Guys	+1.06 yrs	−1.44 – +3.53
Gap, IOP vs Guys	**−5.06 yrs**	−8.66 – −1.34
Error, any group	no difference	all CIs span 0
IOP unseen (extra error)	**+3.30 yrs**	+1.62 – +4.93
HH unseen (extra error)	+0.25 yrs	−0.88 – +1.36

ROBUSTNESS, EXPLANATIONS, UNCERTAINTY

CHECK	RESULT
Site read from features	**91%** (chance 33%)
Scan look alone shifts age	**3–4 yrs**, 100% same direction
IOP offset explained by intensity / blur / fingerprint	no: ≤1 / 0 / persists
Sanity check (randomised layers)	p 0.13, maps destroyed
Deletion test, top vs random 5%	4.83 vs 1.04 yrs
Uncertainty on unseen IOP	doubles, 92% coverage

USAGE GUIDANCE

SUITABLE FOR
- Research on brain ageing within a single acquisition site
- Comparing groups scanned on the same scanner and protocol
- Teaching and methods development

NOT SUITABLE FOR
- Any clinical or individual decision
- Comparing brain-age gaps *across* scanners without site-aware validation
- New sites without checking its uncertainty first

KNOWN LIMITATIONS
- Small subgroups (9 IOP test subjects); single seed and split. The IOP offset nonetheless replicates in validation (−4.1) and test (−5.1).
- In IXI, site is confounded with who each site recruited: the cause of the IOP offset (untested scanner properties or cohort) cannot be separated.
- Healthy volunteers aged 20–86 only. Attribution resolution 8 mm. Uncertainty is the model's own spread, not a calibrated interval.

Evidence: every figure on this page comes from the scripts in **scripts/03b–14** and is logged with dates in **PROJECT_GUIDE.md**. Statistics: age-adjusted OLS with HC3 errors, 5,000-sample bootstrap CIs, FDR where several regions are tested. Wording: acquisition-site effects, not "hospital bias"; no clinical claims.

AUDITED beyond accuracy

The fairness passport.

10 The rules we followed

1. **One model trained on everyone**; results split by group afterwards.
2. Say **"acquisition-site effects"**, never "hospital bias".
3. Never "the model thinks differently"; say **attribution patterns differ**.
4. **No clinical claims**: healthy volunteers, methods audit only.
5. **Control for age** in every comparison.
6. **Report confidence intervals** everywhere.
7. **Never tune on the test set**: it was opened exactly once.

11 Where everything is & how to run it

LOCATION	CONTENTS
GitHub `main`	All code, figures, website, report, passport (passport via pull request #5)
`scripts/`	Our pipeline, numbered in run order `01b` → `16`
`experiments/`	Second SFCN + DeepBrainNet pipeline
Google Drive `BrainAge_Project/`	Checkpoints, all `results/phaseX` folders, website copy
Laptop `D:\ML_Project`	Preprocessed scans (`IXI_preprocessed_v2`), results

To run a step: `python scripts/<script>.py --project-dir <folder>`. Each script's top comment explains its inputs and outputs. Fine-tuning needs a GPU (Colab T4); every other step runs on a laptop CPU in 1–45 minutes.

12 Questions people will ask

QUESTION	SHORT ANSWER
Why SFCN?	Competition winner, pretrained on 14k brains, 3D, and its age-probability output gives uncertainty for free.
Is 3.7 years good?	Yes; typical published range on small datasets is 3–5 years.
Only 9 IOP test people?	Wide range, but it replicates in validation and survives leaving out any one person.
Is IOP's hospital biased?	We call it an acquisition-site effect; in IXI, site and the people recruited are mixed up.
Are the heatmaps trustworthy?	They passed the randomisation and deletion tests.
Did you tune on the test set?	No. Opened once, after every decision was final.
What next?	Same-person multi-scanner data; run the audit on DeepBrainNet; train a scanner-blind model.

THE FINAL SENTENCE TO REMEMBER

Accurate and fair across sex, but not scanner-invariant. Brain-age gaps from it should not be compared across scanners without checking first, and the model's own uncertainty tells you where to be careful.